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AXI Mobility × peaq Milestone Update

What we built, why, how it works, what's next

Partnership update for the peaq tech, DePIN, and partnership team
Pilot context: ROAM Electric — 10 internal EV motorcycles, Nairobi
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Executive summary

AXI Mobility runs a fleet platform. AI agents do the work of monitoring vehicles, drivers, energy, commands, and risk; we turn those signals into cases, actions, and reports. Customer is the supervisor. AXI is the operator.

We built three things on peaq, all live on Agung testnet:

1. **Insurance Risk Oracle** — verifiable risk scores per vehicle and driver. Powers our 5–15% premium-discount pitch to insurers.
2. **High-Risk Command Vault** — K-of-M approval for fleet-wide commands (lock 47 bikes, mass route override).
3. **Credential Revocation Registry** — revocable driver licenses, NTSA inspections, insurance binders. Roadside inspectors and underwriters consult before underwriting.

Plus a TypeScript SDK and standalone CLI that wraps peaq's three pallets (`peaqDid`, `peaqRbac`, `peaqStorage`), peaq EVM via viem, KMS-backed Ed25519 signers, and a daily Merkle anchor service.

Status today. 51 unit and fuzz tests passing locally. Agung deploy dry-run successful. Pre-audit hardening checklist about 80% complete. Production Agung deploy runbook ready (`AGUNG_DEPLOY.md`). Mainnet promote is gated on Safe multisig setup and external audit acceptance.

Why peaq

We chose peaq over generic Layer 1s and Layer 2s for one reason: **mobility-native chain primitives**. Specifically:

What we needed	What peaq gives	Generic-L1 alternative cost
Verifiable identity per vehicle and driver	`peaqDid` pallet (DID method spec, attribute extrinsics)	Build our own DID method and registry contract
On-chain RBAC across operators and regulators	`peaqRbac` pallet	Build OZ-AccessControl-shape registry plus maintain it
Anchor and verify chain-of-custody for compliance	`peaqStorage` plus EVM contracts side-by-side	Pick one; or run two chains
Realistic gas and finality for African DePIN	Substrate finality plus EVM precompiles in a single node	Bridge fees and two-chain ops
Mobility ecosystem alignment	DePIN focus, ROAM-class peers	Generic finance and DeFi peers

We are not bridging across two chains. peaq is the chain. Polygon is our *fallback* for the daily Merkle anchor only — redundancy, not primary. This is a strategic lock, not a tactical call.

What we built (the three use cases)

Each contract solves a real problem we ran into during the ROAM pilot design.

Insurance Risk Oracle

Problem

Insurance underwriters in Kenya quote premiums based on patchy fleet data. They charge a "risk surcharge" because they cannot trust the numbers. ROAM (and every fleet operator we talked to) overpays by 15–25%.

Solution

We attest a per-DID risk score on-chain. Append-only, signed by an authorised attestor. Score in $[0, 1000]$. Insurer reads the latest score before quoting. Verifiable. Auditable. Cheap.

Real example

ROAM rider Mary Akinyi has been riding 6 months, 4,200 km logged, zero harsh-brake events, 89% safety score. AXI attestor signs:
`attest(did:peaq:mary, score=920, expiresAt=now+30d, payloadHash=0x...)`
Insurer reads `getLatest(did:peaq:mary)` and sees a score of 920 $\rightarrow$ premium discount tier 1.

Why peaq vs a centralised database

Insurers will not trust AXI's database; we are an interested party. They will trust an on-chain attestation log signed by independent attestors. The chain gives them deniability ("we relied on a public register").

Implementation

`InsuranceRiskOracle.sol`: UUPS proxy, OpenZeppelin AccessControl, append-only history. Public read; role-gated write. `ATTESTOR_ROLE` for AXI's attestation service; `CALIBRATOR_ROLE` for the score-to-premium curve; `UPGRADER_ROLE` for upgrades; `DEFAULT_ADMIN_ROLE` for role grants.

High-Risk Command Vault

Problem

Fleet-wide actions are risky. "Lock 47 bikes" is one button, and it has to be reversible, auditable, and multi-party. Operator dashboards routinely have a single-sig escape hatch — that is how fleets get held ransom.

Solution

K-of-M approval gate. AXI staff propose a high-risk command (lock fleet, mass route override, broadcast more than 100 messages, fund withdrawal). M approvers must sign. Once K reach (out of M), the command becomes executable by anyone — including a regulator if AXI staff disappear.

Real example

Mary Akinyi's bike (KMG 117P) shows pack temperature 52°C — risk of fire. AXI ops Brian proposes `LOCK_FLEET` on the 3 bikes in the same depot, $K = 2$ of $M = 3$.
Joy approves (1/2 reached). ROAM ops manager approves (2/2 reached). Vault status flips to `Approved`. Anyone calls `execute(commandId, payload)` $\rightarrow$ bikes locked.
The on-chain audit shows: who proposed, who approved, when, and what.

Why peaq vs an off-chain Slack approval

A Slack approval can be repudiated. An on-chain K-of-M cannot. The audit trail is the product — insurance underwriters and regulators want to see that operators are practising good governance before they price the premium or licence the fleet.

Implementation

`HighRiskCommandVault.sol`: *not* upgradeable (an upgradeable approval gate is a back door). OpenZeppelin Access-Control, ReentrancyGuard, Pausable. State machine `Proposed` $\rightarrow$ `Approved` $\rightarrow$ `Executed` (terminal). $1 \leq K \leq M \leq 10$. Replay-safe `commandId` derivation. Permissionless execution once K reached.

Credential Revocation Registry

Problem

Driver licenses, NTSA inspection certificates, and insurance binders all expire or get revoked. Today fleets find out at the roadside check (about $300 fine plus a tow-truck bill). Insurers find out at claim time (about $5,000 loss). Verification is paper-based and manual.

Solution

Each credential is hashed and registered. When the issuer (NTSA, insurance company, AXI fleet manager) revokes, they call `revoke(hash, reasonCode)`. Anyone — insurer, regulator, the driver app itself — can check `isRevoked(hash)` for free. Per-issuer scope: only the issuer of a credential can revoke it.

Real example

ROAM rider Wanjiku's NTSA inspection expires 2026-05-15. NTSA officer revokes:
`revoke(hash=0x... (Wanjiku's certificate), reasonCode=2 (expired))`
At the next checkpoint, the inspector scans the QR code on Wanjiku's bike. The verify app calls `isRevoked(hash)` and gets back `true`. Wanjiku is pulled aside, not jailed for "fraudulent docs". Saves AXI a court date.

Why peaq vs a central NTSA database

NTSA's portal goes down regularly. Our chain does not. Plus: insurance firms outside Kenya want to sell premium products to AXI fleets without trusting Kenya's central databases — chain-anchored revocation gives them an independent verification path.

Implementation

`CredentialRevocationRegistry.sol`: UUPS proxy, OpenZeppelin AccessControl. Idempotent re-revoke (preserves the original `reasonCode`). Batch up to 100 revocations per transaction (handy when an issuer key is rotated). Emergency `pause()` for incidents.

How it all fits together

The data path

```
                                                  +-- Insurance underwriter
                                                  |
                                                  +-- Regulator (NTSA)
                                                  |
ROAM ops dashboard --+                            +-- Financier (lender)
                     |                            |
AXI Atlas Agent -----+--+ Convex Hub  --signs+anchors--> peaq mainnet (3338)
                     |                            |
Driver app ----------+                            +-- Vault (K-of-M)
                                                  +-- Risk Oracle (append-only)
                                                  +-- Revocation Registry
                                                  |
                                                  +-- peaqDid / peaqRbac / peaqStorage
```

Three flows, every day

Flow A — Daily Merkle anchor (00:00 UTC)

1. Convex cron walks yesterday's `auditLogs` (every state change is hash-chained).
2. Builds a Merkle tree (binary, SHA-256 leaves).
3. Calls `anchor(dayId, root)` on **peaq mainnet** primary plus **Polygon mainnet** fallback.
4. `peaqscan`, Subscan, and Polygonscan all show the same root.
5. Anyone can pick a random event from yesterday, walk the proof, and verify.

Flow B — Per-event critical hash (live)

1. ATL agent flags a high-risk event (siphoning, collision, geofence breach in restricted zone).
2. Event is hashed and signed Ed25519 by AXI's KMS-backed signer (the production key never leaves KMS).
3. Tx submitted to peaq mainnet *immediately* (not waiting for the daily anchor).
4. The case in our cloud dashboard shows the chain TX link as a "verified" pill.

Flow C — Public verification (on demand)

Insurance underwriter holds a fleet's risk-attestation hash. They call our `verify.aximobility.com/api/verify/[hash]` endpoint. We return the Merkle proof plus the on-chain TX. They walk the proof against the chain themselves. No trust in AXI required.

The peaq footprint we use

peaq feature	How AXI uses it	Frequency
EVM smart contracts	3 production contracts (Vault, Oracle, Registry)	1–10× per day per fleet
`peaqDid` pallet	DID issuance per vehicle, per driver, per asset (battery, container)	once per entity creation; ~50/day at pilot scale
`peaqRbac` pallet	Cross-tenant role assignments (issuer, attestor, approver, auditor)	rare; once per role change
`peaqStorage` pallet	Selective evidence storage (compliance binders)	sparse; every monthly compliance recompute
EVM precompiles	viem-compatible `defineChain` plus standard tooling	every chain interaction
Mainnet RPC (`peaq.api.onfinality.io`)	Daily anchor plus per-event critical hash	continuous
Agung testnet	Pre-mainnet integration testing	continuous in CI

Pallet vs EVM split (decision rationale)

- **DID + Storage + RBAC** use *pallets* — peaq's native primitives, cheaper, type-safe via Substrate.
- **Vault + Oracle + Registry** use *EVM contracts* — rich state machines plus UUPS upgradability plus audit firms know Solidity.
- **Daily anchor** uses EVM — tooling plus Polygon redundancy on the same artefact.

Milestones reached

Date	Milestone	Status
2026-04	Lock peaq as primary chain (Polygon as fallback) — strategic decision in `POSITIONING.md`	axi-light**Done**
2026-04	Verified 34 peaq facts and flagged 7 open questions in `VERIFIED_SPECS.md`	axi-light**Done**
2026-04	Built TypeScript integration package: EVM client (viem), Substrate client, DID issuer/method/resolver, RBAC + Storage pallet wrappers, KMS-Ed25519 signers, idempotency, retry, structured logger	axi-light**Done**
2026-04	Built standalone `peaq-cli` ops tool with 5 command groups (anchor, did, storage, health, status)	axi-light**Done**
2026-04	Wrote 3 production-shape Solidity contracts with UUPS, OZ AccessControl, custom errors, NatSpec	axi-light**Done**
2026-04	51 unit and fuzz tests passing on local Foundry harness	axi-light**Done**
2026-05	Slither in CI, severity-gated to high; `forge fmt` plus snapshot in CI	axi-light**Done**
2026-05	Agung deploy dry-run successful (chain ID 9990)	axi-light**Done**
2026-05	`AUDIT_CHECKLIST.md` (about 80% items closed)	axi-light**Done**
2026-05	`AGUNG_DEPLOY.md` — 25-minute zero-to-first-attestation runbook	axi-light**Done**
2026-05	Per-contract STRIDE threat models published in `threat-models/`	axi-light**Done**
2026-05	`SECURITY.md` responsible disclosure policy	axi-light**Done**
2026-05	`PEAQ_REVIEW.md` hand-off packet for peaq tech team	axi-light**Done**
2026-05	This milestone document	axi-light**Done**

What's next (the public roadmap)

Milestone	When	What it unlocks
Production Agung deploy	Week 1	Live testing against real Agung; ROAM internal pilot can chain-anchor daily
Internal pilot launch (10 ROAM bikes)	Week 2	First real chain anchor, first real risk attestation, first real revocation
External audit firm engagement	Week 3–4	Lock contracts, tag `audit-v1`
Audit completion plus remediation	Week 4–8	Gate to mainnet
peaq mainnet promote	Month 2	Production daily anchor, Safe 4-of-7 multisig `DEFAULT_ADMIN`
First external customer (REVLOG batteries)	Month 3	Same platform, different ontology
Bug bounty live	Month 3	Phase 5 gate per `EXECUTION_ORDER.md`
First sovereign customer deploy	Month 6+	Atlas Sovereign K2 LoRA on AWS GPU for PII-touching reasoning

What we'd love from the peaq team

Direct answers to seven unverified questions

These come from `VERIFIED_SPECS.md`. We want to publish answers in that file as "verified, source: peaq team confirmation".

1. **Exact extrinsic fees.** What does `peaqDid.addAttribute` cost in PEAQ and USD? What about `peaqStorage.addItem`? We need this for capacity planning and cost-of-anchor model.
2. **DePIN bulk-write pricing.** Is there a tier for high-frequency writes? At ROAM Phase 4 (about 10,000 vehicles), we will write 10–100 events per second.
3. **W3C JSON-LD VC compliance.** Does peaq DID resolve correctly inside W3C VC libraries? Our insurance arbitrage thesis depends on insurers being able to consume our DIDs in their existing tooling.
4. **Decentralisation metrics.** Validators and collators count plus geographic distribution plus nakamoto coefficient. We need this for our threat model and uptime guarantee.
5. **Major incidents and forks since mainnet launch.** For our risk model.
6. **KREST → PEAQ migration plan.** What is the canary network policy? Is there guidance for partners?
7. **Official viem chain definition.** Does peaq plan to ship one? We are synthesising via `defineChain` today.

Code review

- Validate our `did:peaq:...` issuer and resolver against the spec repo.
- Validate our pallet wrapper code (storage and RBAC).
- Spot-check the EVM contracts; flag anything peaq has seen go wrong in other DePIN deployments.

Mainnet promotion guidance

- What is peaq team's preferred audit firm list? (We are considering Trail of Bits, OpenZeppelin, Halborn, Sigma Prime.)
- What monitoring requirements does peaq want to see before a partner goes mainnet?
- Multisig setup recommendations? (Safe on peaq EVM is the path; signers plus threshold.)
- Public launch playbook — does peaq announce and amplify partner mainnet launches? AXI / ROAM is the first east-African EV-fleet integration; it is worth amplifying.

Partnership amplification (when ready)

- Co-marketing for the ROAM mainnet promote — joint announcement with ROAM and a peaq team blog post.
- DePIN ecosystem listing — list AXI Mobility on peaq's DePIN partners page.
- Reference architecture write-up — happy to contribute a "how AXI uses peaq" reference architecture for other DePIN partners.

Next actions

What we need from peaq: a 1-hour kickoff call to walk the contracts, answer the seven questions above, and align on mainnet promotion gates. We have time slots available all week — reply to team@aximobility.com.

Repos for review

- **peaq-contracts**: github.com/aximobility/peaq-contracts
- **peaq-integration**: github.com/aximobility/peaq-integration
- Frozen tag for this review: `peaq-review-2026-05-07`

Documents in this packet

- This document (`AXI_peaq_Milestone_Update.tex`)
- `PEAQ_REVIEW.md` — 13-section technical review packet
- `AUDIT_CHECKLIST.md` — pre-audit hardening status
- `AGUNG_DEPLOY.md` — step-by-step deploy walkthrough
- `SECURITY.md` — responsible disclosure policy
- `threat-models/` — per-contract STRIDE plus DREAD analysis
- `VERIFIED_SPECS.md` — 34 verified facts plus 7 open questions

The shape of the partnership

AXI Mobility brings	peaq brings
ROAM (10 bikes pilot, then hundreds)	Mobility-native chain
REVLOG (battery custody)	DID + RBAC + Storage pallets
Kusini (commodity logistics)	EVM + Substrate single node
KTP (transit data)	DePIN ecosystem
Service-as-Software pricing	Validator network
22 monorepo workspaces	SDK + reference architecture
Atlas AI agent fleet	Mainnet faucet + RPC
9-tool observability stack	Audit firm referrals
Production hardening discipline	Co-marketing reach
	DePIN partner amplification

Both sides ship end of Q2 2026. Both sides win when ROAM goes mainnet on peaq.

Looking forward to the conversation.

Brian Mwai | Joy C. Langat | AXI Mobility Limited | 2026-05-07